



Namal University, Mianwali

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Machine Learning

Introduction to Neural Networks

Neural Networks will take over

- Neural networks have become one of the main approaches to AI
- They have been successfully applied to various pattern recognition, prediction, and analysis problems
- In many problems they have established the state of the art
 - Often exceeding previous benchmarks by large margins
 - Sometimes solving problems, you couldn't solve using earlier ML methods

Some of the Major Breakthroughs

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Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

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Image: Adobe Stock

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.



Most Popular Newsletters News Alerts

How do you feel about Black Friday and Cyber Monday?

- ☐ They're great -- I get a lot of bargains!
- ☐ The deals are too spread out -- I'd prefer just one day.
- ☐ They're a fun way to kick off the holiday season.
- ☐ I don't like the commercialization of Thanksgiving Day.
- ☐ They're crucial for the retail industry and the economy.
- ☐ The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

AWS Enjoys Top Perch in IaaS, PaaS Markets

US Comptroller Gears Up for Blockchain and

Some of the Major Breakthroughs



The Keyword

Latest Stories

Product News

Topics



TRANSLATE

NOV 15, 2016

Found in translation: More accurate, fluent sentences in Google Translate

Barak Turovsky

PRODUCT LEAD, GOOGLE TRANSLATE

In 10 years, Google Translate has gone from supporting just a few languages to 103, connecting strangers, reaching across language barriers and even helping



Some of the Major Breakthroughs



Some of the Major Breakthroughs

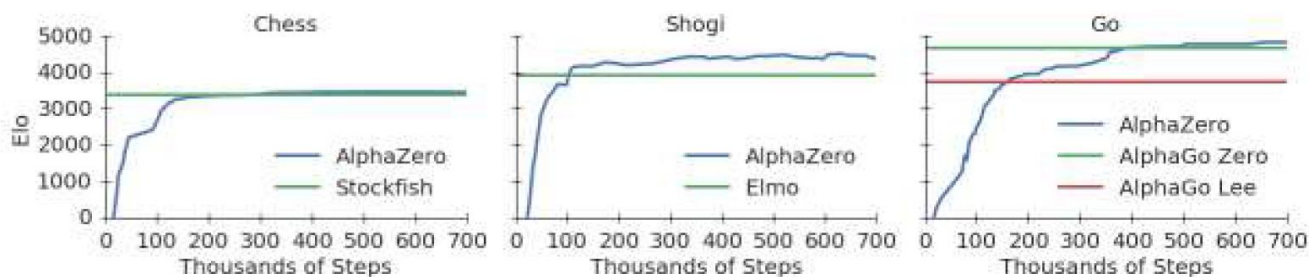
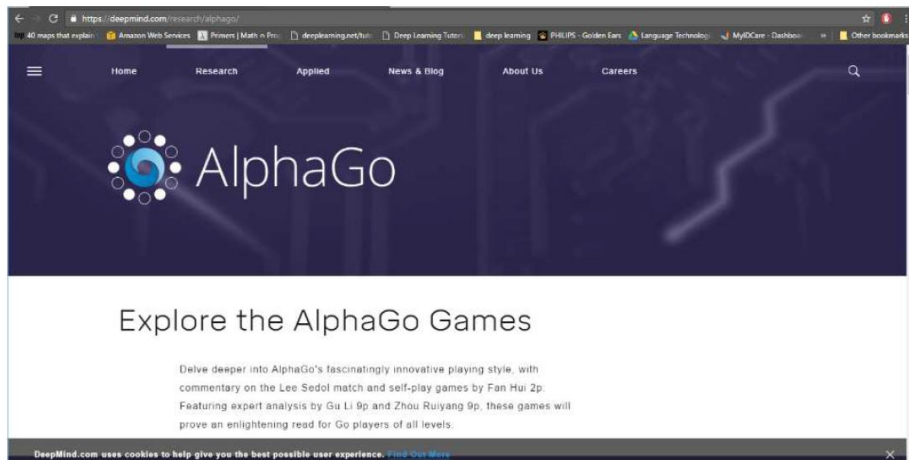


Figure 1: Training *AlphaZero* for 700,000 steps. Elo ratings were computed from evaluation games between different players when given one second per move. **a** Performance of *AlphaZero* in chess, compared to 2016 TCEC world-champion program *Stockfish*. **b** Performance of *AlphaZero* in shogi, compared to 2017 CSA world-champion program *Elmo*. **c** Performance of *AlphaZero* in Go, compared to *AlphaGo Lee* and *AlphaGo Zero* (20 block / 3 day) (29).

Some of the Major Breakthroughs



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."



"young girl in pink shirt is swinging on swing."



"man in blue wetsuit is surfing on wave."

What are Neural Networks?

- It all starts here..



The amazing Capacity of Humans

- Humans can
 - Learn
 - Solve problems
 - Recognize patterns
 - Create
 - Reflect
- Worthy of emulation?

Interesting thought..

“If the brain was simple enough to be understood - we would be too simple to understand it!”

– Marvin Minsky

Early Theory for Human Cognition

Humans learn through association

Lightning is generally followed by thunder

- “hey here’s a bolt of lightning, we’re going to hear thunder”
- “We just heard thunder; did someone get hit by lightning”?

Association

- “Pairs of thoughts become associated based on the organism’s past experience”
- Learning is a mental process that forms associations between temporally related phenomena

Aristotle's four laws of Association

- The law of contiguity

Things or events that occur close together in space or time get linked together

- The law of frequency

The more often two things or events are linked, the more powerful that association.

- The law of similarity

If two things are similar, the thought of one will trigger the thought of the other

- The law of contrast

Seeing or recalling something may trigger the recollection of something opposite.

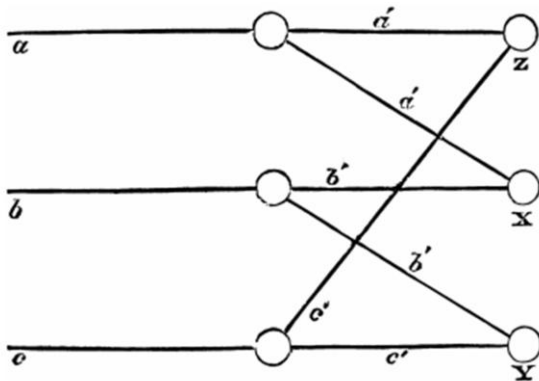
But where are these associations stored?

The start of connectionism!

The information is in the connections (1873)

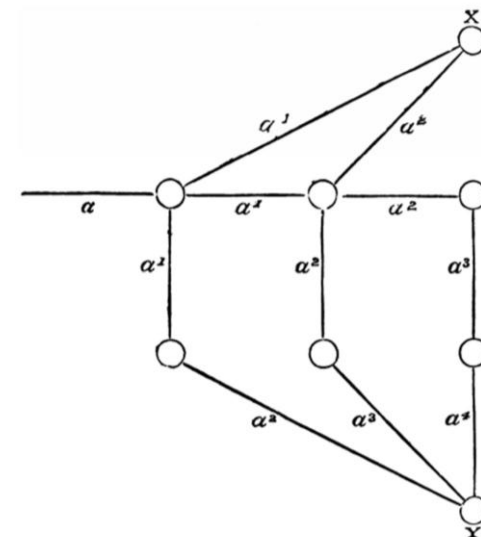
-Alexander Bain

- The brain is a mass of interconnected neurons



- Different combinations of inputs can result in different outputs

Different intensities of activation of A lead to the differences in when X and Y are activated



Alexander Bain's Assumptions

In 1873, Bain postulated that there must be one million neurons and 5 billion connections

- In 1883, Bain was concerned that he hadn't taken into account the number of "partially formed associations" and the number of neurons responsible for recall/learning
- By the end of his life (1903), recanted all his ideas!
 - Too complex; the brain would need too many neurons and connections

Quick Recap

Neural network-based AI has taken over most AI tasks

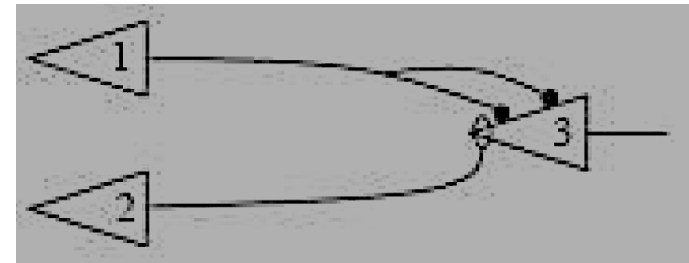
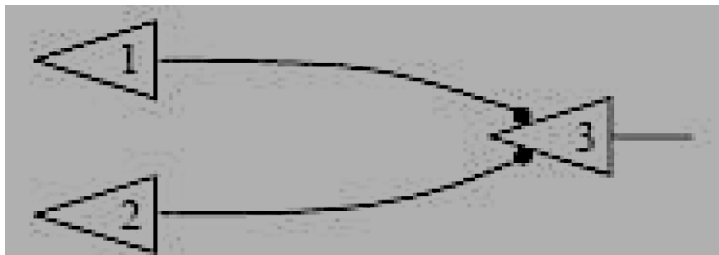
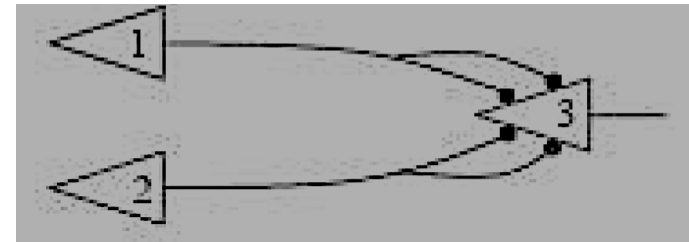
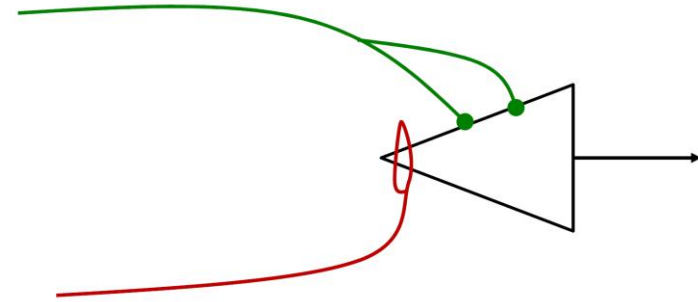
- The earliest model of cognition was associationism
- The more recent model of the brain is connectionist
 - Neurons connect to neurons
 - The workings of the brain are encoded in these connections
- Current neural network models are connectionist systems with following requirements
 - The connectivity of units
 - The activation function of units
 - The nature of the learning procedure that modifies the connections between units

What is the unit?

The Unit

Excitation Input: Transmits input to the neuron

- **Inhibition Input:** Any signal to the inhibition input prevents neuron from firing



Hebbian Learning Mechanism

Organization of Behavior, 1949. - Donald Hebb

– “When a cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A’s efficiency, as one of the cells firing B, is increased.”

- As A repeatedly activates B, its ability to activate B improves.

In mathematical terms: $w_{xy} = w_{xy} + \eta xy$

Weight of the connection from input neuron **X** to output neuron **Y**.

- This simple formula is actually the basis of many learning algorithms in ML

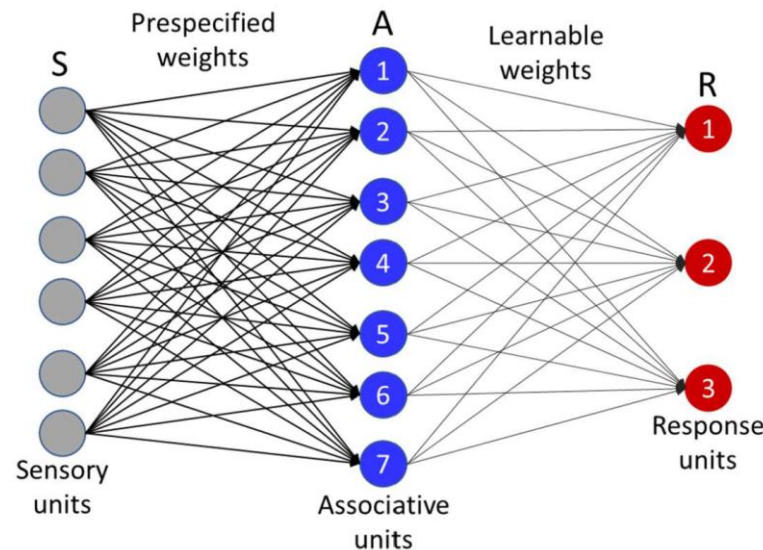
An Improved Model

Frank Rosenblatt

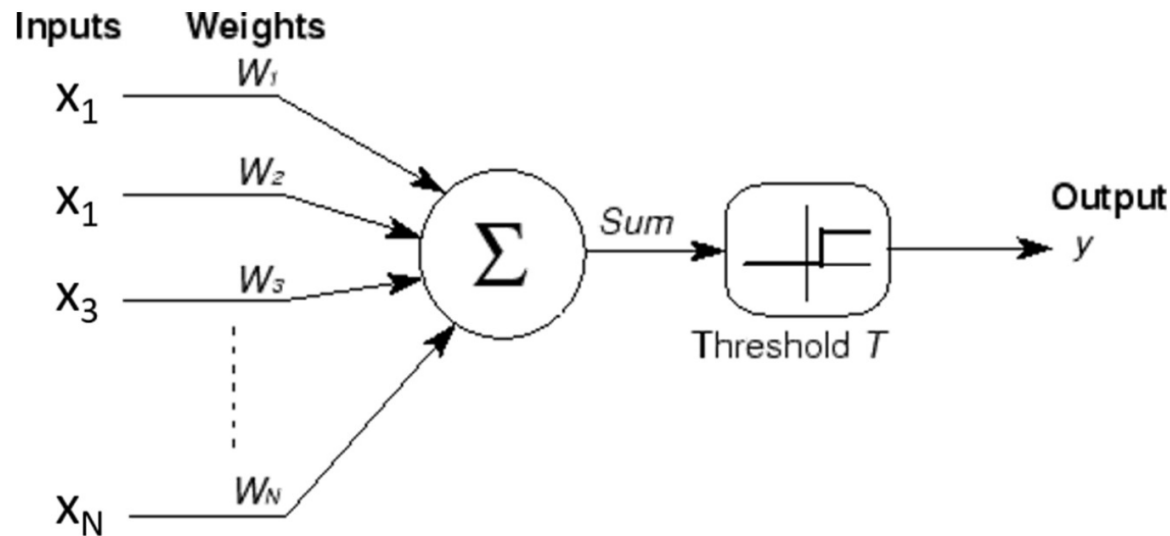
- Inventor of the solution to everything, aka the Perceptron (1958)

Simplified perceptron model

- Association units combine sensory input with fixed weights
- Response units combine associative units with learnable weights



Perceptron



- Number of inputs combine linearly
 - Threshold logic: Fire if combined input exceeds threshold

$$Y = \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

Perceptron

$$\mathbf{w} = \mathbf{w} + \eta(d(\mathbf{x}) - y(\mathbf{x}))\mathbf{x}$$

Sequential Learning:

$d(x)$ is the desired output in response to input $\mathbf{x}$

$y(x)$ is the actual output in response to $\mathbf{x}$

- Boolean tasks
- Update the weights whenever the perceptron output is wrong
 - Update the weight by the product of the input and the *error* between the desired and actual outputs
- Proved convergence for linearly separable classes

Thank you